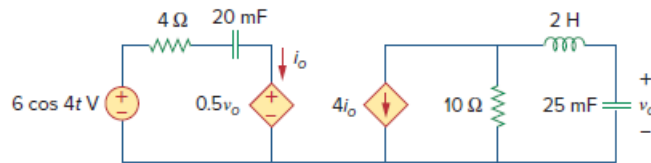


Problem

Use *PSpice* or *MultiSim* to find v_o and i_o in the circuit of Fig. 10.130 below.

Figure 10.130:



Step-by-step solution

Step 1 of 3

The schematic is shown below. We insert I_{PRINT} & PRINT to print

I₀ & V₀ in the output file. Since $\omega = 4$, $f = \frac{\omega}{2\pi} = 0.6366$,

We set Total pts = 1, Start frequency = 0.6366 and End frequency = 0.6366. In the AC sweep box. After simulation, the output file includes,

<i>Frequency</i>	<i>VM(\$N_0002)</i>	<i>VP(&N_0002)</i>
$6.366E^{-01}$	$3.496E^{+01}$	$1.261E^{+01}$
<i>Frequency</i>	<i>1M(V_PRINT2)</i>	<i>IP(V_PRINT2)</i>
$6.366E^{-01}$	$8.912E^{-01}$	$-8.870E^{+01}$

Step 2 of 3

Therefore,

$$\begin{aligned}
 V_o &= 34.96 \angle 12.6^\circ V, \\
 I_o &= 0.8912 \angle -88.7^\circ A, \\
 V_o &= 34.96 \cos(4t + 12.6^\circ) V \text{ \& } \\
 i_o &= 0.8912 \cos(4t - 88.7^\circ) A.
 \end{aligned}$$

Step 3 of 3

